

# MULTIPARTITE ENTANGLEMENT IN 1D HAF MODEL

*A thesis Submitted*  
in Partial Fulfilment of the Requirements  
for the Degree of  
**MASTER OF SCIENCE**

*by*  
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*to the*

**School of Physical Sciences**

**National Institute of Science Education and Research**

**Bhubaneswar**

**Date**

# MULTIPARTITE ENTANGLEMENT IN 1D HAF MODEL

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## **DEDICATION**

I dedicate this thesis to my parents, my teachers and my past self.

## DECLARATION

I hereby declare that I am the sole author of this thesis in partial fulfillment of the requirements for a postgraduate degree from National Institute of Science Education and Research (NISER). I authorize NISER to lend this thesis to other institutions or individuals for the purpose of scholarly research.

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Signature of the Student

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The thesis work reported in the thesis entitled "Multipartite Entanglement in 1D HAF Model" was carried out under my supervision, in the school of Physical Sciences at NISER, Bhubaneswar, India.



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## ACKNOWLEDGEMENTS

I sincerely thank my supervisor, Dr. Anamitra Mukherjee, for our insightful and extensive discussions and his perspective. I am fortunate to have had the opportunity to work closely and learn from him, and will cherish this enriching experience. Furthermore, I would like to thank my colleagues Aritra Mukhopadhyay and Subhajyoti Pal for their significant help with computation and numerical analysis, and Dr. Pradeep Thakur for his insights about the role of entanglement in studying condensed matter systems. Lastly, I thank the many-body group for the several enriching discussions in condensed matter physics and beyond.

# Abstract

Entanglement is a quantum phenomenon central to modern quantum information science and the study of many-body quantum systems. This thesis explores the detection and quantification of entanglement, particularly in multipartite settings, using both traditional and information-theoretic tools. We review foundational separability criteria, introduce and formulate the notion of entanglement witnesses<sup>131,244</sup>, and explore their limitations in various settings (asymptotic regime, mixed states, multipartite depth et cetera). Then, we discuss the concept, formulation and evaluation of entanglement measures, including concurrence and tangle<sup>51,282</sup>, and introduce the quantum Fisher information (QFI) as a thermodynamically accessible measure for multipartite entanglement<sup>205</sup>. We utilize these measures to study the entanglement in the 1D isotropic Heisenberg antiferromagnetic spin- $\frac{1}{2}$  model, and discuss their usefulness and limitations. QFI is observed to be specially effective and experimentally feasible. Analytical and numerical calculations of QFI for the target model enables extensive characterization of multipartite entanglement in the system, demonstrating its scaling behavior (with temperature, system size and magnetic field) and correspondence with known separability bounds. Furthermore, we extend its use to the XY model for multiparameter estimation, highlighting its potential in phase diagnostics. This work supports QFI as a versatile and experimentally feasible tool for entanglement detection in strongly correlated systems.

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